

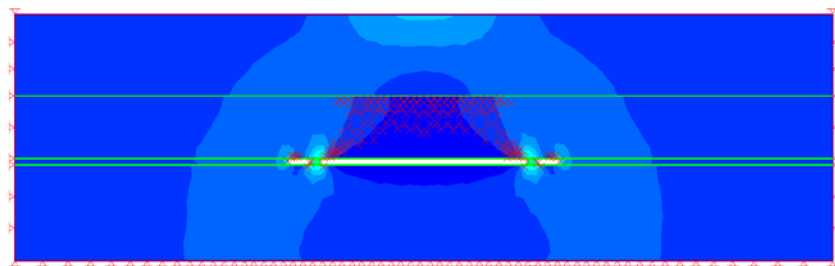
Development of a Stope in a Coal Mine

This tutorial will model the development of a stope and access roads in a coal mine. The tutorial demonstrates the use of multiple materials and staging in *Phase2*, including specification of different in-situ stresses for each material layer.

The complete model can be found in the **Tutorial 28 Coal Mine Stope.fez** file. All tutorial files installed with *Phase2* 8.0 can be accessed by selecting File > Recent Folders > Tutorials Folder from the *Phase2* main menu.

Topics covered

- Staged excavation
- Multiple materials
- Custom field stresses
- Locked-in horizontal stress
- Advanced discretization
- Copy boundary



Model

Start the *Phase2* Model program by double-clicking on the *Phase2* icon in your installation folder.

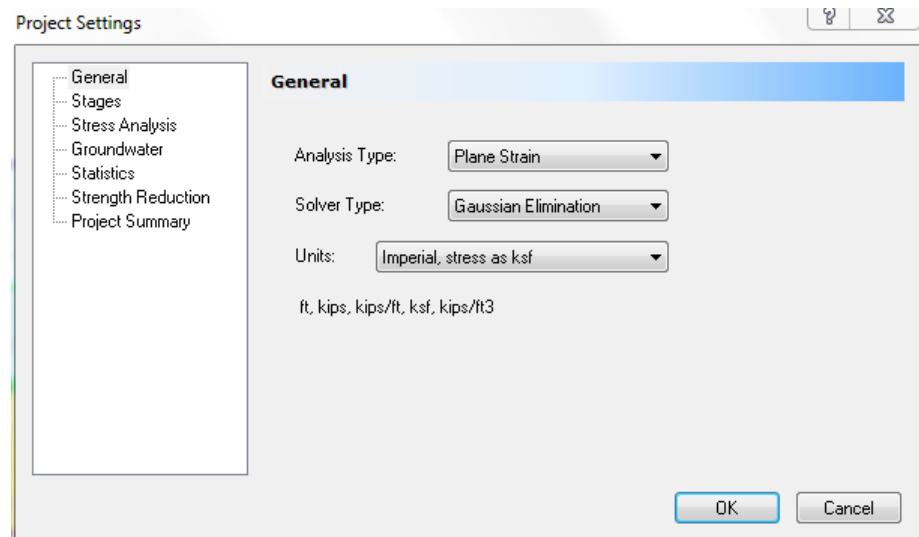
Project Settings

For staged models, the first thing we must always do is to set the Number of Stages in the Project Settings dialog.

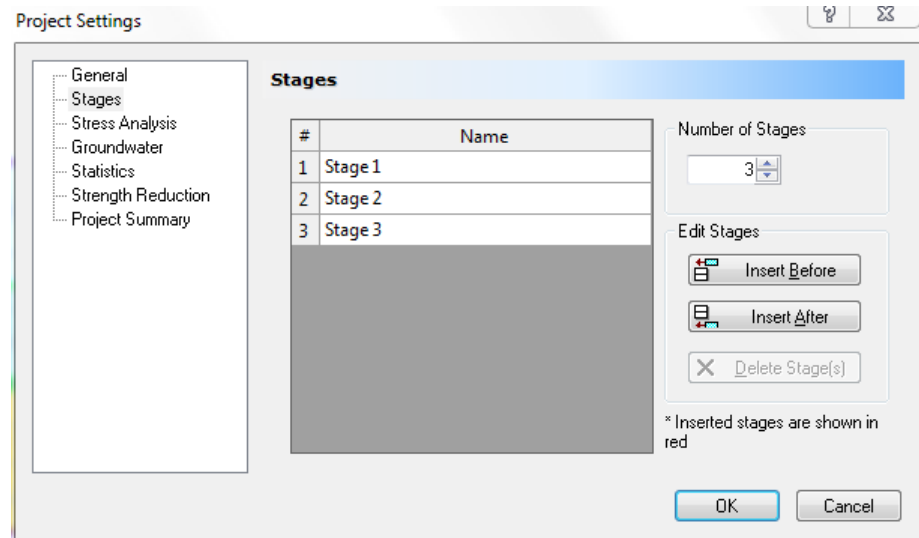


Select: Analysis → Project Settings

With the General tab selected, set the Units to “Imperial, stress as ksf” as shown. Note that *Phase2* remembers the most recently selected Units in Project Settings, and uses it as the default setting for all new documents.



Select the Stages tab. Change the Number of Stages to 3 as shown.



Select OK to close the dialog.

Entering Boundaries

We will enter the external boundary for the model.



Select: Boundaries → Add External

Enter the following information in the command (prompt) line at the bottom right corner of the *Phase2* Model window.

```
Enter vertex[t=table,i=circle,esc=cancel]: 0 0
Enter vertex[...]: 2000 0
Enter vertex[...]: 2000 597
Enter vertex[...]: 0 597
Enter vertex[... ,c=close,esc=cancel]: c
```

Press F2 to Zoom All.

We will now add the material boundaries, which will define the rock mass layers.



Select: Boundaries → Add Material

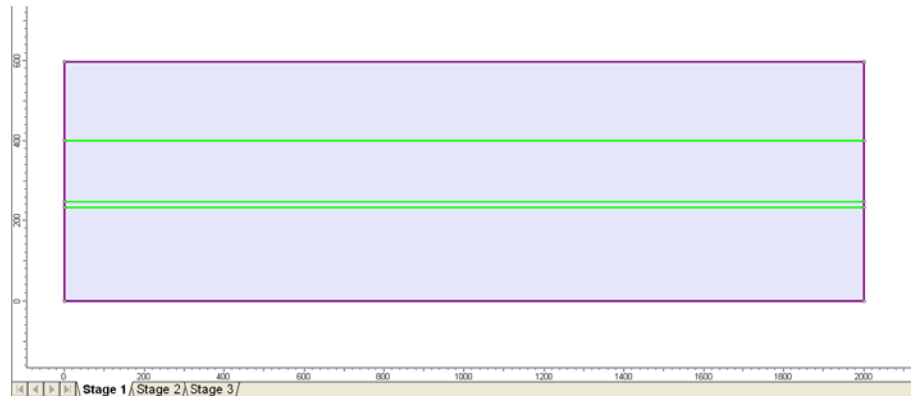
```
Enter vertex [t=table,i=circle,esc=cancel]: 2000 400
Enter vertex [...]: 0 400 press Enter
```

Because the material boundaries in our model are all parallel, we can specify the other boundaries by copying the first boundary.

Select: Boundaries → Edit → Copy

Click on the material boundary to select it and hit Enter. We are now prompted to select one of the vertices of the material boundary and specify its new location. Select the vertex with coordinates (2000, 400), then specify its new location as (2000, 247) and hit Enter. The new boundary is created.

To specify the last material boundary, we will repeat the copy procedure. Right-click the mouse on the previously created material boundary and choose Copy Boundary from the resulting pop-up menu. Select the vertex at (2000, 247) and then enter the new location as (2000, 232). The model should now look like this:



We will now add stage boundaries to define the location of the stope and access roads within the coal seam.



Select: Boundaries → Add Stage

Enter vertex [t=table,i=circle,esc=cancel]: **1330 232**
Enter vertex [...]: **1330 247** press Enter

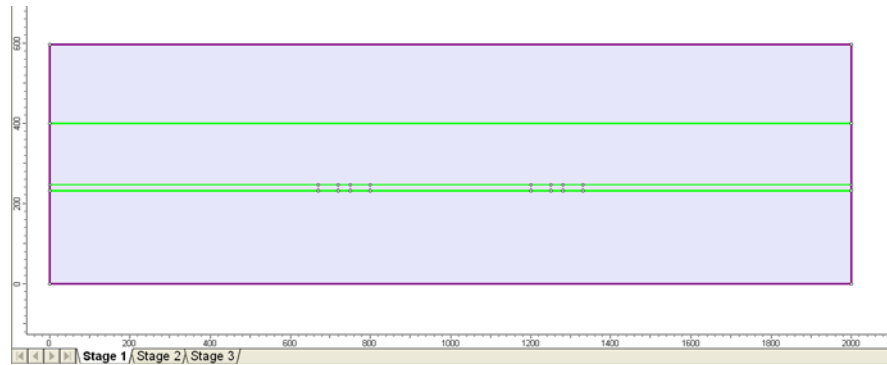
Select the stage boundary that was just created by right-clicking on it with the mouse and choosing Copy Boundary (you may need to zoom in to do this. Zoom in by spinning the wheel on your mouse). Select the vertex with coordinates (1330, 232). Next enter the location (1280, 232) to which it should be copied.

These two boundaries delineate one of four access roads. We will generate three more access roads by copying the two boundaries we just created and placing them at the appropriate locations.

Select: Boundaries → Edit → Copy

Select both stage boundaries and press Enter. Select vertex (1330, 232) and then enter its new location as (1250, 232).

Repeat the procedure two more times, using new locations of (800, 232) and (720, 232). The model should look like this:



This completes the definition of boundaries for this model.

Meshing

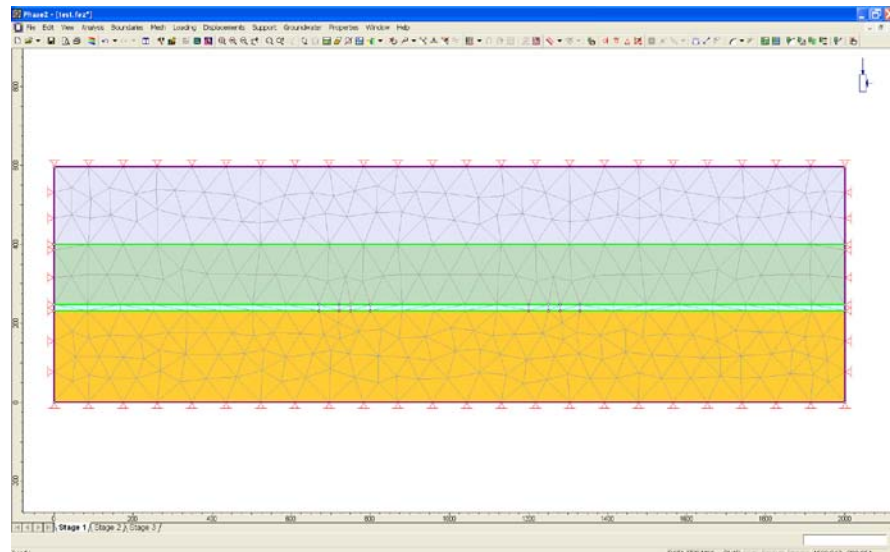
For this model, we will initially use the default Mesh Setup parameters. Select the **Discretize and Mesh** shortcut. A single mouse click on this button automatically discretizes the boundaries and generates the mesh.



Select: Mesh → Discretize & Mesh

The mesh is generated (see the image below) and the status bar will show the total number of elements and nodes in the mesh. You should get the number of nodes and elements indicated below.

ELEMENTS = 476 NODES = 270



This mesh needs to be refined in the area of the pillars. To do so, we will use the Advanced Mesh Regions option in the Mesh Setup dialog.



Choose **Mesh Setup** from the **Mesh** menu. Click the Advanced button and select the Use Advanced Mesh Regions checkbox.

To create a new mesh region select the **Add** button. The mouse cursor turns into a cross with which we will be able to draw our window of interest. Move the mouse to a location with coordinates close to (600, 320), press and hold the left mouse button, and start dragging the mouse to a second point at approximately (1400, 150). A window is drawn on the screen once the mouse begins to move. Release the mouse button once the mouse gets to the indicated coordinate. In the Mesh Setup dialog, change the Element Length to 7.5 as shown.

Mesh Setup ? ▲ ✕

Mesh and Discretization Settings

Mesh Type: Graded ▼

Element Type: 3 Noded Triangles ▼

Gradation Factor: 0.1

Default Number of Nodes on External: 60

Advanced ▼

☒ Use Advanced Mesh Regions

Region	Mesh Type	Element Length
1	Graded	7.5

Add... ▼

Delete ▼

Reset ▼

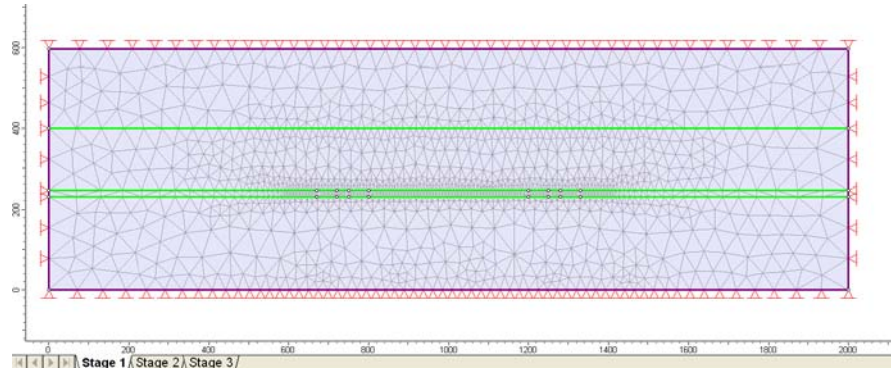
☒ Show Mesh Regions as Hatched Polygons

☐ Improve Discretization Grading

☒ Do Not Discretize Arc or Circle Segments

Discretize Mesh OK Cancel

Press the "Discretize" button followed by "Mesh" button. Click OK to close the dialog. Your mesh should look similar to the following.



Now the mesh is refined and has more elements in the area of interest.

Boundary Conditions

Now we can set the boundary conditions. The portion of the external boundary representing the ground surface (0, 597 to 2000, 597) must be free to move in any direction.



1. Select the **Free** option in the **Displacements** menu.
2. Use the mouse to select the line segment defining the ground surface.
3. Right-click and select Done Selection.

TIP: you can also right-click on a boundary to define its boundary conditions.

The surface is now free, however, this process has also freed the vertices at the upper left and upper right corners of the model. Since these edges should be restrained, we have to make sure that these two corners are restrained.

Let's use the right-click shortcut to assign boundary conditions:

1. Right-click the mouse directly on the vertex at (0,597). From the popup menu select the Restrain X,Y option.
2. Right-click the mouse directly on the vertex at (2000,597). From the popup menu select the Restrain X,Y option.

The displacement boundary conditions are now correctly applied.

Define Material Properties

We will now specify the properties of the different materials in the model.



Select: Properties → Define Materials

With the first material selected at the top left of the Define Material Properties dialog, create a Sandstone material and enter the following properties:

The screenshot shows the 'Define Material Properties' dialog box for a material named 'Sandstone'. The left pane lists materials from Sandstone to Material 29. The right pane shows the following settings:

- Name:** Sandstone
- Material Color:** (light blue)
- Initial Element Loading:** Field Stress & Body Force
- Unit Weight: (kips/ft3):** 0.175
- Elastic Properties:**
 - Elastic Type:** Isotropic
 - Poisson's Ratio:** 0.3
 - Young's Modulus (ksf):** 50000
 - Young's Modulus (resid) (ksf):** 417709
 - E1 (ksf):** 417709, **E2 (ksf):** 417709, **Ez (ksf):** 417709
 - v12:** 0.2, **v1z:** 0.2, **v2z:** 0.2
- Strength Parameters:**
 - Failure Criterion:** Mohr Coulomb
 - Material Type:** Plastic
 - Tensile Strength (peak) (ksf):** 30, **Dilation Angle (deg):** 0
 - Fric. Angle (peak) (deg):** 38, **Fric. Angle (resid) (deg):** 38
 - Cohesion (peak) (ksf):** 62, **Cohesion (resid) (ksf):** 62
 - Tensile Strength (resid) (ksf):** 0
- Stage Properties:** (unchecked)
- Datum Dependent:** (unchecked)
- Unsaturation Shear Strength:**
 - Phi b:** 0, **Air Entry (ksf):** 0

Buttons at the bottom include 'Copy To...', 'Statistics...', 'Show only properties used in model', 'OK', and 'Cancel'.

Select the second, third and fourth materials, respectively, and enter properties for a Siltstone, Coal and Shale as indicated in the next three screen captures. Select OK when you are finished.

Define Material Properties

Siltstone

Name: Siltstone Material Color: 

Initial Element Loading: Field Stress & Body Force Unit Weight: (kips/ft3): 0.17

Elastic Properties

Elastic Type: Isotropic Poisson's Ratio: 0.3

Young's Modulus (ksf): 38000 ☐ Young's Modulus (resid) (ksf): 417709

E1 (ksf): 417709 E2 (ksf): 417709 Ez (ksf): 417709

v12: 0.2 v13: 0.2 v23: 0.2

Strength Parameters

Failure Criterion: Mohr Coulomb  Material Type: Plastic

Tensile Strength (peak) (ksf): 20 Dilation Angle (deg): 0

Fric. Angle (peak) (deg): 28 Fric. Angle (resid) (deg): 28

Cohesion (peak) (ksf): 30 Cohesion (resid) (ksf): 30

Tensile Strength (resid) (ksf): 0

☐ Stage Properties ☐ Datum Dependent

Define Factors... Define Properties...

Unsaturated Shear Strength

Phi b: 0 Air Entry (ksf): 0

Copy To... Statistics... ☐ Show only properties used in model OK Cancel

Define Material Properties

Coal

Name: Coal Material Color: 

Initial Element Loading: Field Stress & Body Force Unit Weight: (kips/ft3): 0.1719

Elastic Properties

Elastic Type: Isotropic Poisson's Ratio: 0.3

Young's Modulus (ksf): 30000 ☐ Young's Modulus (resid) (ksf): 417709

E1 (ksf): 417709 E2 (ksf): 417709 Ez (ksf): 417709

v12: 0.2 v13: 0.2 v23: 0.2

Strength Parameters

Failure Criterion: Generalized Hoek-Brown  Material Type: Plastic

Intact Comp. Strength (ksf): 500 Dilation Parameter: 0

mb Parameter (peak): 2 mb Parameter (resid): 2

s Parameter (peak): 0.052 s Parameter (resid): 0.052

a Parameter (peak): 0.6 a Parameter (resid): 0.6

☐ Stage Properties ☐ Datum Dependent

Define Factors... Define Properties...

Unsaturated Shear Strength

Phi b: 0 Air Entry (ksf): 0

Copy To... Statistics... ☐ Show only properties used in model OK Cancel

Define Material Properties

Shale

Name: Shale Material Color:

Initial Element Loading: Field Stress & Body Force Unit Weight: (kips/ft3): 0.172

Elastic Properties

Elastic Type: Isotropic Poisson's Ratio: 0.3

Young's Modulus (ksf): 45000 ☐ Young's Modulus (resid) (ksf): 417709

E1 (ksf): 417709 E2 (ksf): 417709 Ez (ksf): 417709

v12: 0.2 v13: 0.2 v23: 0.2

Strength Parameters

Failure Criterion: Mohr Coulomb Material Type: Plastic

Tensile Strength (peak) (ksf): 15 Dilation Angle (deg): 0

Fric. Angle (peak) (deg): 38 Fric. Angle (resid) (deg): 38

Cohesion (peak) (ksf): 60 Cohesion (resid) (ksf): 60

Tensile Strength (resid) (ksf): 0

☐ Stage Properties ☐ Datum Dependent

Unsaturated Shear Strength

Phi b: 0 Air Entry (ksf): 0

☐ Show only properties used in model

This completes the definition of material properties.

Field Stress

Now define the in-situ stress field.



1. Select the **Field Stress** option in the **Loading** menu.
2. Change the Field Stress Type from Constant to Gravity (gravitational stress distribution throughout the model).
3. Select the **Use actual ground surface** checkbox. By using this option, the program will automatically determine the ground surface elevation above every finite element and define its vertical stress based on the weight of material overlying it.
4. We wish to specify a different horizontal field stress for each material. Click on the "Advanced" button in the dialog. Tabs with the names of the materials in our model appear. This option enables us to further customize the in-situ stress field. Select the **Apply Custom Field Stress** checkbox.
5. The default horizontal to vertical stress ratios of 1 indicate a hydrostatic initial stress state. For our example, we will change the ratios to 0.428 for the four materials in the model. So enter 0.428 for the two Stress Ratios for each material. Note that these values override the stress ratios of 1 shown in the main dialog.

6. Due to previous tectonic stress history, we will apply locked-in horizontal stresses for each layer of material. Use the following table to enter tectonic in-plane and out-of-plane horizontal stresses for the four materials.

Material	Locked-in-horizontal stresses
Sandstone	120
Siltstone	90
Coal	50
Shale	70

The dialog for sandstone should appear as shown

Field Stress Properties

Field Stress Type: Gravity

☒ Use actual ground surface ☐ Use effective stress ratio ☐ Use variable stress ratio

Ground Surface Elevation (ft): 0

Unit Weight of Overburden (kips/ft³): 0.1719

Total Stress Ratio (horiz/vert in plane): 1

Total Stress Ratio (horiz/vert out-of-plane): 1

Locked-in horizontal stress (in plane) (ksf, Comp. +): 0

Locked-in horizontal stress (out-of-plane) (ksf, Comp. +): 0

OK Cancel Statistics... Advanced <<

Sandstone

☒ Apply Custom Field Stress

Ground Surface Elevation (ft): 0

Unit Weight of Overburden (kips/ft³): 0.1719

Total Stress Ratio (horiz/vert in plane): 0.428

Total Stress Ratio (horiz/vert out-of-plane): 0.428

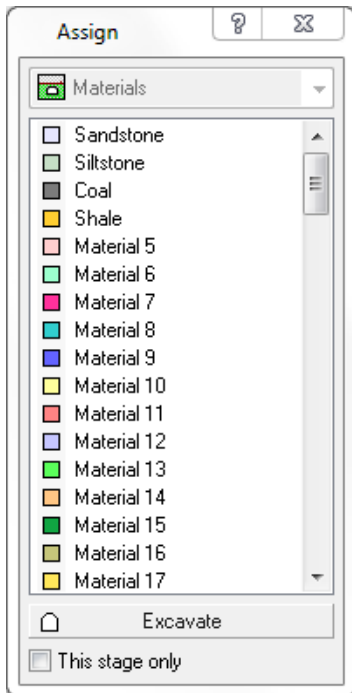
Locked-in horizontal stress (in plane) (ksf, Comp. +): 120

Locked-in horizontal stress (out-of-plane) (ksf, Comp. +): 120

Material Selection List:

- ☐ Sandstone
- ☐ Siltstone
- ☐ Coal
- ☐ Shale
- ☐ Material 5
- ☐ Material 6
- ☐ Material 7
- ☐ Material 8
- ☐ Material 9
- ☐ Material 10
- ☐ Material 11
- ☐ Material 12
- ☐ Material 13

We are finished defining the material properties. Select OK to close the Define Material Properties dialog.



Stage 1 – material assignment

Assigning Properties

Select: Properties → Assign Properties

The Assign Properties dialog allows you to specify the stratigraphy of material layers in our model. In conjunction with the Stage Tabs at the bottom left of the view, it also allows us to assign the sequence in which openings are excavated and supports are installed.

- At the first stage we will only have the material stratigraphy.
- At the second stage we will excavate the road gates on both sides of the slope.
- At the third stage we will excavate the slope.

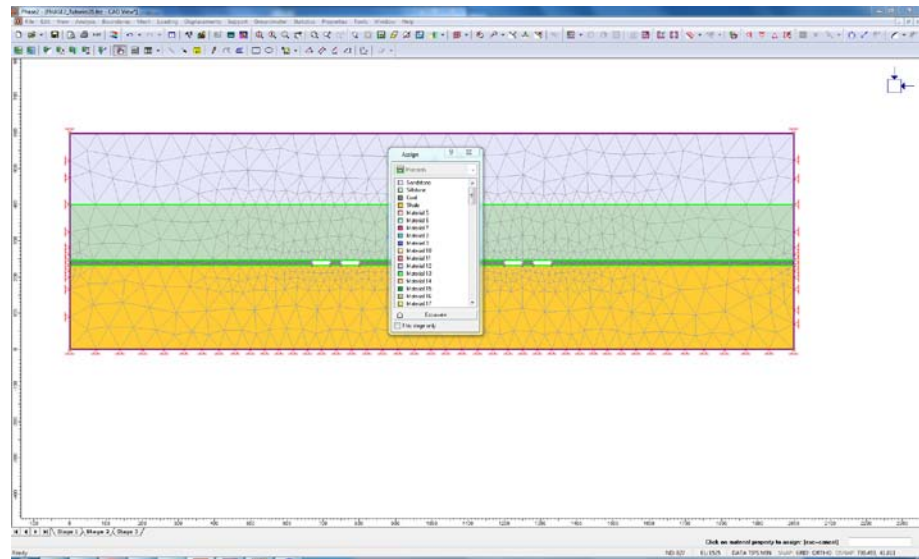
Assign Materials

1. Make sure the Stage 1 tab is selected (at the bottom left of the view).
2. Select the “Siltstone” button in the Assign dialog. (Notice that the material names you entered when you defined the four materials appear on the list).
3. Click the left mouse button in the **second** layer from the top. Notice that these elements are now filled with the colour representing the Siltstone material.
4. Assign "Coal" and "Shale" to the third and fourth layers, respectively. For the coal layer, you will need to click inside each section of the layer since it is separated by stage boundaries.

NOTE: Since we defined the “Sandstone” properties using the *first* tab in the Define Materials dialog, the “Sandstone” properties do not need to be assigned by the user. The properties of the first material in the Define Materials dialog are always automatically assigned to all elements of the model.

Stage 2 – excavation of four access roads

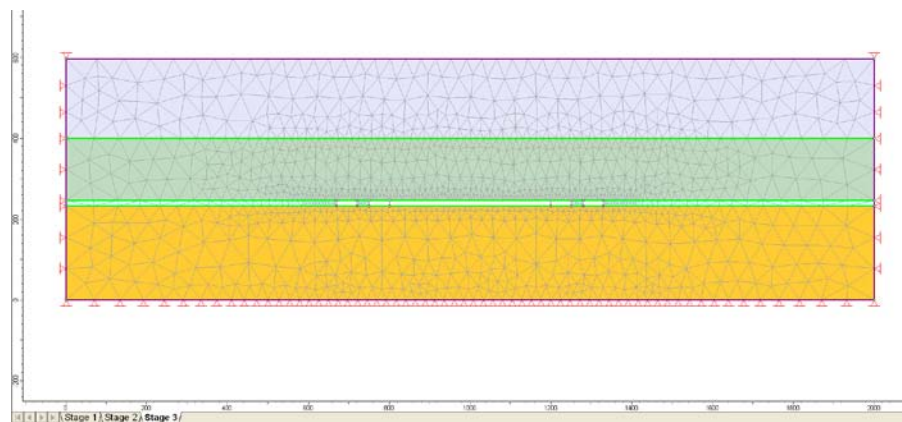
5. Select the Stage 2 tab.
6. Select the “Excavate” button in the Assign dialog.
7. Place the cursor in each of the four access roads, and left click to excavate them as shown (zoom in if necessary).



**Stage 3 – excavation of
slope**

8. Select the Stage 3 tab.
9. Place the cursor in the middle section of the coal layer and click the left mouse button, and the elements will disappear.
10. You are finished assigning materials. Now select each Stage Tab, starting at Stage 1, and verify that the excavation staging and material property assignment is correct.

You have now completed the model building phase of the analysis; the model should appear as in the following figure for Stage 3.



Before you analyze your model, save it as **coalstope.fez**.



Select: File → Save

Use the Save As dialog to save the file. You are now ready to run the analysis.



Select: Analysis → Compute

The *Phase2* Compute engine will perform the required finite element calculations. The analysis should take less than a minute.

Once the computations are done, we will view the results in Interpret.

Interpret

To view the results of the analysis:



Select: Analysis → Interpret

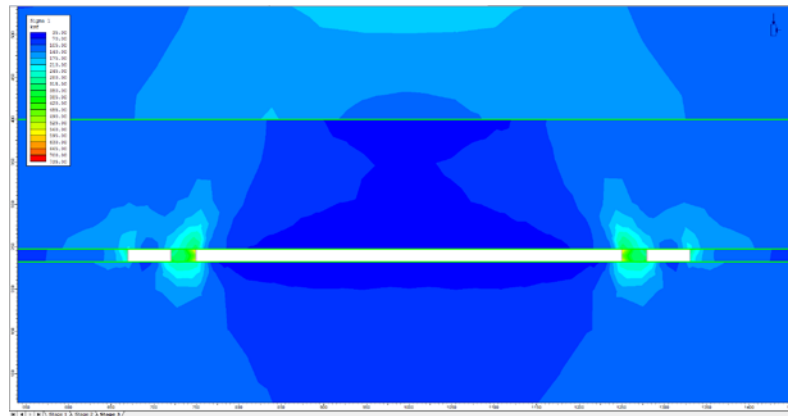
This will start the *Phase2* Interpret program.

Sigma 1

You are now viewing Sigma 1 (major principal) stress contours for Stage 1. You may want to zoom in on the excavations (use the middle mouse wheel or select **Zoom Window** and draw a window around the region of interest). Select the Stage 2 and 3 tabs and observe the changes in stress distribution.

TIP: you can also use the Page Up / Page Down keys to change between stages.

The figure below shows Sigma 1 for Stage 3. Note the high stress in the pillars on either side of the stope.



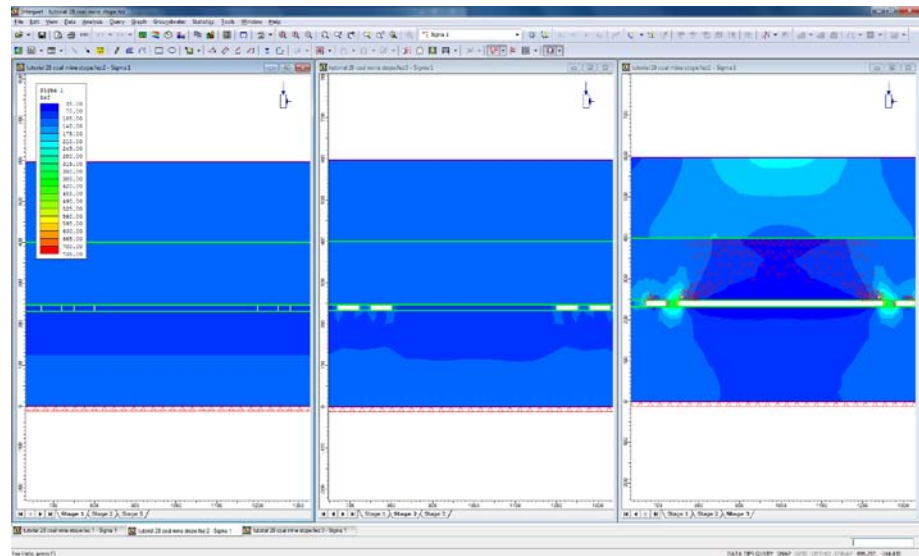
Toggle on the principal stress trajectories, using the button provided in the toolbar. Again select the stage tabs 1 to 3, and observe the stress flow around the excavation.

If you want to compare results at different stages on the same screen, it can easily be done as follows.

1. Select Window→New Window TWICE, to create two new views of the model.
2. Select the Tile Vertically button in the toolbar, to tile the three views vertically.
3. Zoom in on the excavation in each view.
4. Select the Stage 1 tab in the left view, the Stage 2 tab in the middle view, and the Stage 3 tab in the right view.
5. Display the stress trajectories in each view.
6. Hide the legend in the right and middle views (use View → Legend Options, or right-click on a Legend and select Hide Legend).
7. Right-click in any view and select Contour Options. Click in each view, and select Auto-Range (all stages), to ensure that the same contour range is used for all stages. Close the Contour Options dialog.



Your screen should appear as shown below.



Strength Factor

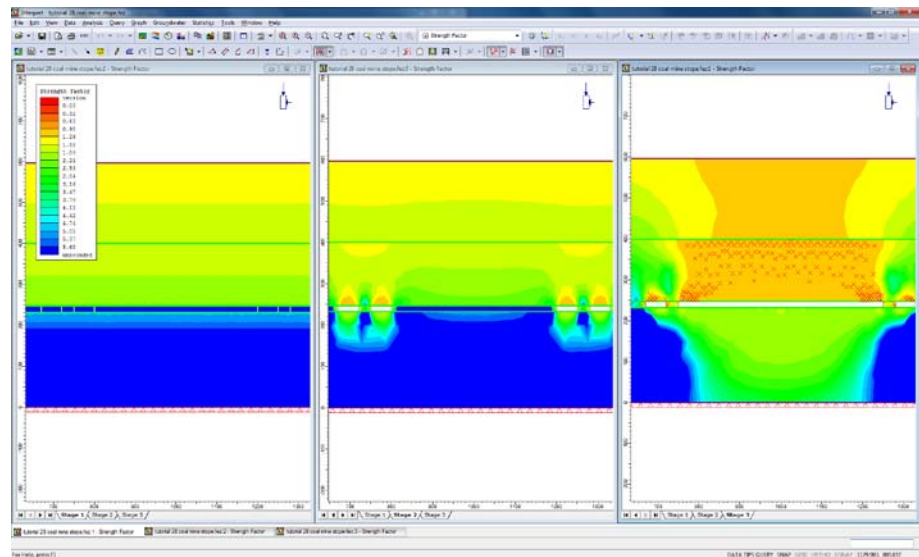
Let's look at the Strength Factor contours.

1. Display the Strength Factor in each view.
2. Turn OFF the stress trajectories and Toggle Yielded Elements ON, using the Display toolbar buttons, in each view.



Observe the development of strength factor and yielding around the excavation. Note that the yielding is in the roof of the stope.

Note: the strength factor is the rock strength divided by the rock stress. For a plastic analysis such as this, the strength factor is always greater than or equal to 1, but the strength factor contours can indicate regions that are close to failure (the lower the value, the closer to failure). See the help files for more information.



Let's view the model full screen again. Maximize one of the views (it doesn't matter which one). Re-display the legend if necessary (View → Legend Options), and select the Stage 3 tab, if necessary.

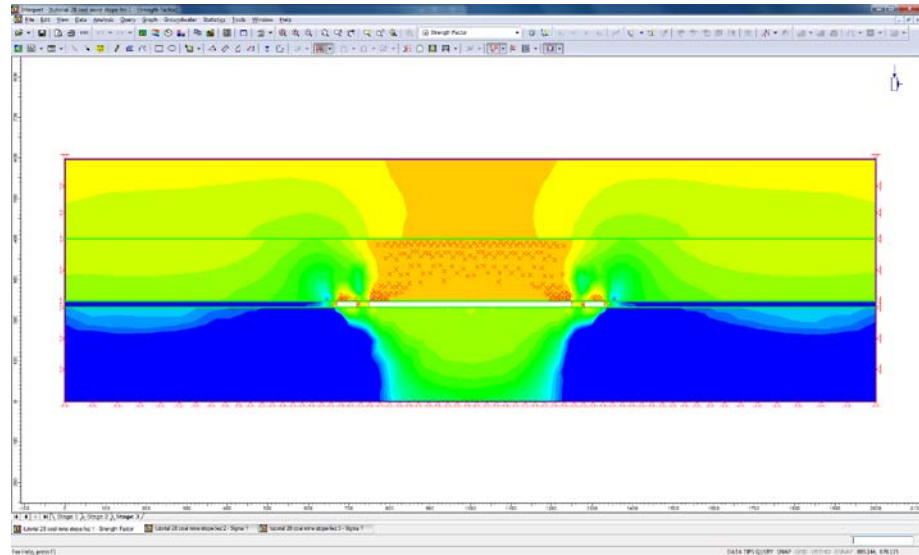


Display the mesh by selecting the Elements button in the toolbar. Note that each Yielded Element symbol corresponds to a single finite element.

As an optional step, use the arrow keys (up / down / left / right), to pan the model around the view. View the contours and yielded elements around the entire excavation. You can also pan by holding down the mouse wheel and moving the mouse.



Toggle off the Mesh and select Zoom All.



Strength Factor contours and yielded elements, Stage 3.

Select the Stage tabs 1 to 3, and observe the strength factor contours on the whole model.

Toggle off the Yielded Elements.

Total Displacement

Plot contours of total displacement by selecting Total Displacement from the pull-down menu in the toolbar. Select the Stage 1 tab. The maximum total displacement for Stage 1 is virtually 0, as indicated in the status bar.

Maximum Total Displacement = 4.81431e-012 ft
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Select the Stage 2 tab.

Maximum Total Displacement = 0.0846708 ft

Select the Stage 3 tab.

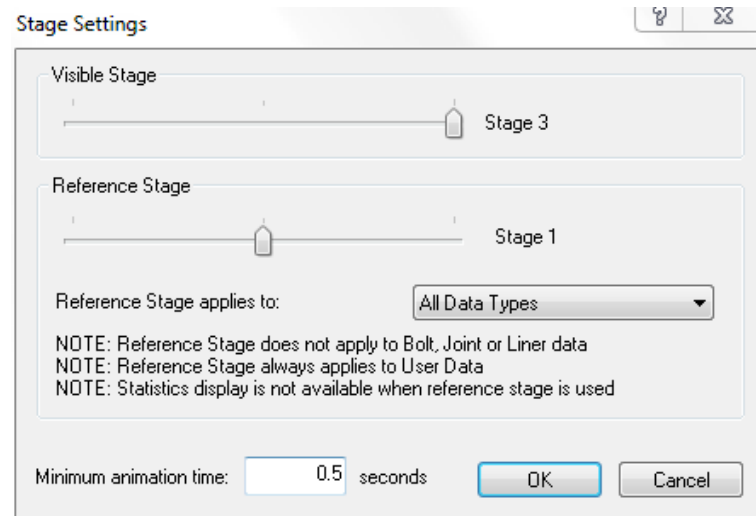
Maximum Total Displacement = 1.41 ft

You can see how the displacement increases as excavation proceeds. (You might see slightly different numbers in your model due to different meshing). As you can see from the displacement contours, the maximum total displacement at Stage 3 occurs at the center of the stope roof.

Differential Results

So far, we have always used a Reference Stage = 0. Differential results *between any two stages* can be viewed by setting the Reference Stage > 0 in the Stage Settings dialog. For example:

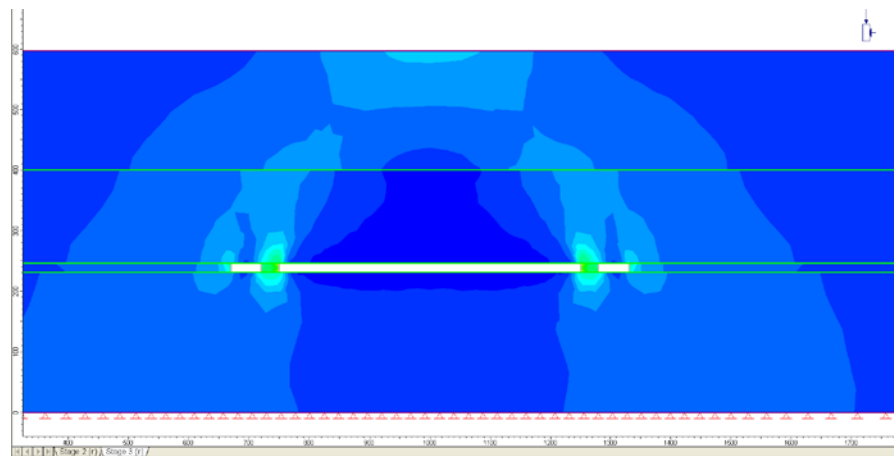
Select: Data → Stage Settings



Set the Reference Stage to 1 and select OK.

Notice that the Stage Tabs now allow you to view results relative to the reference stage you just entered.

If you change the contours back to Sigma 1, you will now see the stresses due to excavation only – the plots are relative to Stage 1, so the gravity stresses are subtracted out. The plot below shows Sigma 1 in Stage 3 relative to Stage 1. See the *Phase2* Help system for more information about how to interpret differential results.



This concludes the tutorial.